CSE260

Assignment 03

This assignment must be hand-written. Show ALL steps in ALL questions

1. Design a circuit diagram for the following system that takes two 4-bit binary numbers [A and B] as inputs and outputs in the following fashion:
   - If A is odd and B is divisible by 10, the output should be **A+B**
   - For other cases, the output should be **A-B**

2. **[Ungraded]** Design a **13 people** attendance system using necessary **parallel and half adder(s)**. Your circuit diagram should be efficient, i.e. use the least amount of components.

3. Design a **32 people** voter system using necessary **parallel and full adder(s)**. Your circuit diagram should be efficient, i.e. use the least amount of components.

4. Design a **BCD to Excess-9** system using necessary parallel adder(s).

5. Construct the function $F(a,b,c,d,e) = \sum(0, 1, 2, 4, 7, 9, 12, 14)$:
   Using 3x8 decoder(s) and 2x4 decoder(s) (you need to use both in the same circuit)
   Your circuit must be efficient, which means you need to use the least amount of components.

6. Construct the function $F(a,b,c,d,e) = \sum(0,1, 5, 8, 9, 13, 16, 19, 24, 25, 31)$:
   a. Using 16x1 and 2x1 mux(s) (you need to use both in the same circuit)
   b. Using 8x1 and 4x1 mux(s) (you need to use both in the same circuit)

   Your circuit must be efficient, which means you need to use the least amount of components.

7. Construct the following functions using a single 4x1 mux:
   a. **[Ungraded]** $F(a,b,c) = \sum(1, 3, 4, 5, 7)$
   b. $F(a,b,c,d) = \sum(0, 2, 3, 7, 8, 9, 10, 11, 15)$

8. You have to construct the truth table and internal circuit diagram for a **4x2 priority encoder.** The priority is given in the following manner:
   a. If the number of 1 in a binary input is divisible by 3, then priority will be given to the MSB line.
   b. If the number of 1 in a binary input is divisible by 2, then priority will be given to the LSB line.
   c. For all other cases, the priority will be given to the MSB line.

9. **[Ungraded]** Design a 16x1 mux using only a **single 4x1 mux**. You may use other basic gates as required.